11. Sequences and Series

11.7. Strategy for Testing Series

11.7 Learning Objectives
- Summarize the different techniques for determining convergence and divergence in series.

First we have a few known series that we can quickly determine the convergence of.

Theorem 11.7.1 **Geometric Series Test**

For the series $\sum_{n=1}^{\infty} a \cdot r^{n-1}$ we have the following.

If $|r| < 1$, the **geometric series** $\sum_{n=1}^{\infty} ar^{n-1}$ converges, and

$$\sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r}$$

r is called the **ratio**

If $|r| \geq 1$ the series diverges.

Theorem 11.7.2 **P-Test for p-series**

For a p-series of the form $\sum_{n=1}^{\infty} \frac{1}{n^p}$ we have the following.

1. If $p = 1$, we call the series

$$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \ldots$$

   the Harmonic Series. It diverges.
2. If $p > 1$, the series converges.
3. If $p < 1$, the series diverges.

Then we get into the rest of the tests for series convergence.

Theorem 11.7.3 **Divergence Theorem (Test for Divergence)**

If the series $\sum_{n=1}^{\infty} a_n$ is convergent, then $\lim_{n\to\infty} a_n = 0$.

Therefore,
- If $\lim_{n\to\infty} a_n \neq 0$, then $\sum_{n=1}^{\infty} a_n$ is divergent.
- If $\lim_{n\to\infty} a_n = 0$, then the series may converge or diverge. The test is inconclusive.

Pros	Cons	Notes
Usually an easy limit to compute	if $\lim_{n\to\infty} a_n = 0$, we cannot conclude anything	you may need to use l'Hospital's rule, remember to do this with functions of x

Theorem 11.7.4 The Integral Test

Let $(a_n)_{n=1}^{\infty}$ be a sequence of positive terms. Suppose that $a_n = f(n)$, where f is a continuous, positive, and decreasing function of n for all $n \geq N$ (where N is a positive integer). Then,

(i) If $\int_N^{\infty} f(x)\,\mathrm{d}x$ is convergent, then $\sum_{n=1}^{\infty} a_n$ is a convergent series.

(ii) If $\int_N^{\infty} f(x)\,\mathrm{d}x$ is divergent, then $\sum_{n=1}^{\infty} a_n$ is a divergent series.

Pros	Cons	Notes
We can use integral techniques!	Lots of conditions/hypotheses to check, and we have to compute an improper integral.	This is usually a "last resort" if other tests fail. Also, remember limit notation for improper integrals. Often this is useful if there are logarithms in your series.

Theorem 11.7.5 Alternating Series Test

If an alternating series, $\sum_{n=1}^{\infty} a_n$ of the form

$$\sum_{n=1}^{\infty} (-1)^{n-1} b_n \quad \text{or} \quad \sum_{n=1}^{\infty} (-1)^{n} b_n,$$

where $b_n \geq 0$ satisfies:

(i) $b_{n+1} \leq b_n$ for all $n \geq N$

(ii) $\lim_{n\to\infty} b_n = 0$

then the series $\sum_{n=1}^{\infty} a_n$ is convergent.

Pros	Cons	Notes
Great for alternating series, not much else	Does not tell you divergence, use divergence test for this...	If (i) does not hold, try ratio or root test.

Theorem 11.7.6 The Direct Comparison Test (Comparison Test)

Suppose that $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are series with positive terms.

(i) If $\sum_{n=1}^{\infty} b_n$ is convergent and $a_n \leq b_n$ for all $n \geq N$, then $\sum_{n=1}^{\infty} a_n$ converges.

(ii) If $\sum_{n=1}^{\infty} b_n$ is divergent and $a_n \geq b_n$ for all $n \geq N$, then $\sum_{n=1}^{\infty} a_n$ diverges.

Pros	Cons	Notes
Usually easy if you can find b_n.	Have to show and inequality, and also figure out b_n	Usually you choose $\sum b_n$ to be a geometric or p series.

Theorem 11.7.7 The Limit Comparison Test

Suppose that $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are series with positive terms. If

$$\lim_{n\to\infty} \frac{a_n}{b_n} = c$$

where c is a finite positive number ($c > 0$), then either both seris converge, or both series diverge.

Pros	Cons	Notes
Similar to DCT, but no longer need to show an inequality.	Sometiems the limit we need to compute has tricky algebra, be careful.	Same as DCT, choose $\sum b_n$ carefully, probably geometric or p series.

Theorem 11.7.8 The Ratio Test

For a series $\sum_{n=1}^{\infty} a_n$

(i) If $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = L < 1$, then the series $\sum_{n=1}^{\infty} a_n$ is **absolutely convergent** (and therefore convergent).

(ii) If $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = L > 1$, then the series $\sum_{n=1}^{\infty} a_n$ is **divergent**.

(iii) If $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = L = 1$, then the Ratio Test is **inconclusive**. The series $\sum_{n=1}^{\infty} a_n$ may absolutely converge, conditionally converge, or diverge. We cannot determine this with ratio test.

Pros	Cons	Notes
No conditions to check!	If $L = 1$, need to try something else	Works well when we have factorials and exponents.

Theorem 11.7.9 The Root Test

(i) If $\lim_{n\to\infty} \sqrt[n]{|a_n|} = L < 1$, then the series $\sum_{n=1}^{\infty} a_n$ is **absolutely convergent** (and therefore convergent).

(ii) If $\lim_{n\to\infty} \sqrt[n]{|a_n|} = L > 1$, then the series $\sum_{n=1}^{\infty} a_n$ is **divergent**.

(iii) If $\lim_{n\to\infty} \sqrt[n]{|a_n|} = L = 1$, the Root Test is **inconclusive**. i.e. $\sum_{n=1}^{\infty} a_n$ may absolutely converge, conditionally converge, or diverge. We need a different test to figure out which.

Pros	Cons	Notes
No conditions to check!	If $L = 1$, need to use another test.	Works well when $a_n =$ (something with $n)^n$

Exercise 11.7.1

Take the time to make flash cards and/or a flowchart for the different series convergence tests above.

11.7 Section Summary:

- We summarized all the different **series** convergence tests and when you might use them.